

Document Image Rectification using Stable Diffusion Transformer

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Abstract

Document images captured using handheld devices often suffer from geometric distortions caused by perspective variations, warping, and lens-induced aberrations. These distortions negatively impact text readability, OCR accuracy, and automated document analysis, making effective rectification essential. Traditional approaches, such as 3D reconstruction-based flattening and convolutional neural network (CNN)-based warping prediction, have shown promising results but struggle with handling complex, non-uniform distortions and long-range dependencies in document structures. In this paper, we propose a novel Conditional Stable Diffusion Transformer based framework designed specifically for document image rectification. Unlike conventional UNet-based diffusion models, which rely on hierarchical convolutional operations, our transformer-based architecture provides a global receptive field through self-attention mechanisms, enabling precise structural preservation and text alignment. Furthermore, we incorporate cross-attention conditioning, allowing the model to integrate auxiliary information for improved rectification accuracy. To enhance efficiency and robustness, we introduce a coarse rectification using control points and thin plate spline that estimates an initial globally aligned structure before the diffusion-based refinement process. Extensive experiments on benchmark datasets demonstrate that our approach achieves state-of-the-art rectification accuracy while maintaining comparable inference time to existing deep learning-based solutions. Our proposed framework establishes a new paradigm for document image rectification by leveraging transformer-based modeling, generative diffusion processes, and conditional guidance, making it highly effective across a wide range of document distortions.

1. Introduction

With the increasing reliance on smartphones for document digitization, capturing high-quality document images has become essential for various applications, including opti-

cal character recognition (OCR), automated text analysis, document retrieval, and digital archiving. However, images captured using handheld devices often suffer from geometric distortions caused by factors such as camera angle variations, document warping, and lens-induced perspective distortions. These distortions can significantly degrade document readability, leading to errors in downstream tasks such as text extraction and information retrieval. Addressing this issue has led to extensive research in document image rectification, a process aimed at restoring distorted document images to their original undistorted form. Traditional methods for document image rectification have relied on 3D reconstruction techniques, where the deformed document surface is estimated and subsequently flattened. These approaches typically involve multi-view imaging, structured lighting, or laser scanning to recover the 3D geometry of the document. While effective, such methods require specialized hardware, making them impractical for widespread use in consumer applications. Recent advancements in deep learning have led to the development of data-driven rectification approaches, where convolutional neural networks (CNNs) and transformer-based architectures are trained to predict warping functions that map distorted images to rectified outputs.

In this paper, we have proposed a Conditional Stable Diffusion Transformer based framework that addresses the lim-

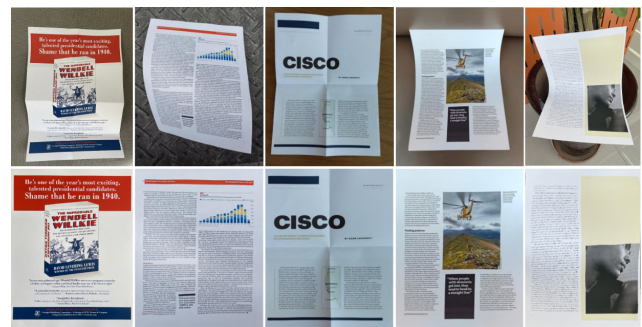


Figure 1. Qualitative results of our proposed model on sample images. The first row depicts distorted input document images, while the second row presents the corresponding rectified outputs.

itations of traditional CNN-based approaches. Unlike conventional UNet-based diffusion models, which rely on convolutional operations for feature extraction and denoising, our proposed transformer-based diffusion architecture enables the network to effectively capture long-range dependencies and global context within a document image. The self-attention mechanism allows the model to focus on key structural elements, preserving the document’s layout and textual integrity throughout the rectification process. A key advantage of our approach lies in its ability to condition the diffusion process using auxiliary rectification information through cross-attention mechanisms. Instead of concatenating feature maps at multiple scales, as done in traditional UNet-based models, our transformer network directly integrates conditional embeddings into the rectification process. This leads to better alignment, improved text legibility, and enhanced geometric consistency across diverse document types as seen in Figure 1. Furthermore, we leverage control points and edge-based features to guide the initial document transformation before applying the final diffusion-based refinement. Our work introduces several key advancements in the field of document image rectification.

- We have proposed a transformer-driven diffusion based rectification model that overcomes the limitations of existing architectures by leveraging self-attention mechanisms for long-range dependency modeling.
- Our approach introduces cross-attention-based conditioning, allowing the rectification process to incorporate background segmentation, control point estimation, and structural priors, leading to more accurate results.
- By combining transformer-based modeling, diffusion-driven refinement, and cross-attention conditioning, our proposed method establishes a new paradigm for document image rectification, offering significant improvements in accuracy, generalization, and adaptability across diverse document types.
- Extensive experiments on multiple benchmark datasets demonstrate that our approach achieves state-of-the-art performance for document image rectification.

2. Related Works

Document image rectification is a crucial task in computer vision and document processing, aiming to correct distortions in scanned or captured document images to improve readability and downstream applications such as Optical Character Recognition (OCR). Existing approaches to document rectification can be broadly categorized into 3D shape reconstruction-based methods and deep learning-based approaches. Early works in document rectification leveraged 3D shape reconstruction techniques, often requiring additional hardware or structured lighting to estimate the document’s surface geometry. Brown and Seales [2] pioneered the use of illumination-based methods to recover a

document’s surface shape, employing a mass-spring particle model for flattening. Similarly, Zhang et al. [29] introduced a laser range scanner-based approach to reconstruct the document’s 3D structure, which was then corrected using physical modeling techniques. Other studies explored multi-view reconstruction, where multiple images captured from different viewpoints were used to infer the document’s 3D shape. For example, Tsoi et al. [22] developed a multi-view approach that leveraged stereo imaging for rectification. An alternative to hardware-based reconstruction is single-image shape estimation, where the 3D shape of the document is inferred from a single distorted image. Courteille et al. [3] employed shape-from-shading techniques to estimate surface curvatures and restore the document to a planar form. While these methods were effective in controlled settings, they often required calibrated cameras, known light sources, or extensive preprocessing, making them less practical for real-world applications.

Recent advances in deep learning have led to data-driven approaches that directly learn distortion patterns from large-scale document datasets. One of the first deep learning-based methods, DocUNet [17], introduced a fully convolutional network to predict pixel-wise displacement fields, enabling the restoration of distorted documents without explicit 3D reconstruction. Later, Li et al. [15] proposed a patch-based rectification technique, where documents were divided into local patches, rectified individually, and then stitched together to form the final output. The introduction of DewarpNet [5] further improved document rectification by incorporating 3D shape learning within a UNet-based architecture, providing better generalization across diverse document layouts. More recently, transformer-based approaches have been explored to capture long-range dependencies and global context. Feng et al. [7] introduced Doctr, leveraging vision transformers (ViTs) to learn structural relationships within document images. The DocTr++ model [10] extended this idea with a hierarchical encoder-decoder framework, refining rectification by incorporating multi-scale feature representations.

Other recent methods, such as DocReal [27], have incorporated attention mechanisms and control point estimation, enhancing the rectification of local deformations and background noise elimination. These deep learning-based approaches have demonstrated significant improvements over traditional 3D reconstruction techniques, particularly in handling unconstrained distortions such as folds, warping, and perspective distortions.

3. Basic Idea

Document image rectification aims to correct distortions present in scanned or photographed document images, ensuring a geometrically accurate and visually coherent representation. Traditional methods often rely on geometric

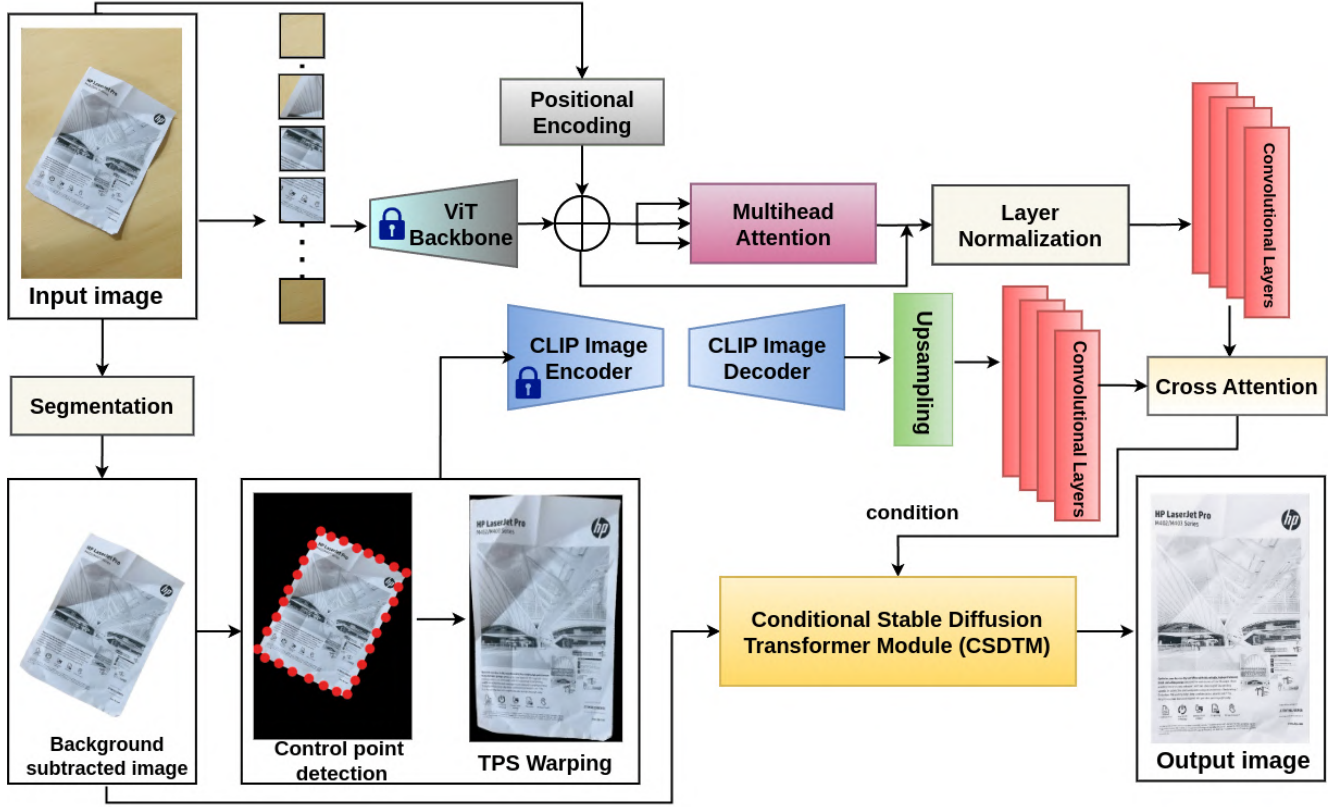


Figure 2. Overview of the proposed methodology for document image rectification, illustrating the key components and workflow of our approach.

priors, 3D reconstruction, or deep neural networks to model and correct distortions. However, these approaches struggle with handling diverse real-world distortions such as folds, warping, and perspective deformations.

In this work, we have proposed a Stable Diffusion Transformer based architecture for document image rectification. Our method leverages the generative power of diffusion models combined with the structured reasoning capabilities of transformers. By formulating rectification as a progressive denoising and transformation process, we aim to iteratively refine the document’s structure while maintaining semantic consistency. Diffusion models operate by progressively adding noise to an image and learning to reverse this process to recover the original structure. Given a distorted document image x_0 , forward noising is defined as,

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I) \quad (1)$$

where, α_t is a noise schedule parameter that controls the variance at each step. The goal is to learn a reverse denoising function $p(x_{t-1}|x_t)$ that reconstructs the undistorted document step-by-step:

$$p(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 I) \quad (2)$$

where, $\mu_\theta(x_t, t)$ is a learned denoising function parameterized by a neural network. To effectively guide this denoising process for rectification, we integrate a Transformer-based encoder-decoder that captures structural dependencies within the document. We have employed a vision transformer (ViT)-based encoder E that extracts hierarchical document features from the distorted image representation:

$$F = E(x_t) \quad (3)$$

where, F represents multi-scale feature embeddings that capture global and local deformations. The decoder network D utilizes these features to guide the diffusion model in reconstructing a rectified version of the document:

$$x_0 = D(F, x_t) \quad (4)$$

We further refine rectification by conditioning the diffusion process. This ensures that rectified documents retain textual integrity while improving structural consistency. The final output is obtained through iterative refinement:

$$x_{t-1} = x_t - \eta \nabla_{x_t} L(x_t, F) \quad (5)$$

where, $L(x_t, F)$ is the loss function enforcing geometric and perceptual consistency, and η is the learning rate. Our Conditional Stable Diffusion Transformer Module (CSDT) effectively combines the generative capabilities of diffusion models with the structured understanding of transformers to rectify document images in a data-driven manner. Unlike previous approaches, CSDT does not require explicit geometric priors, making it more adaptable to real-world document rectification scenarios. The experimental results demonstrate the robustness of SDT in handling complex distortions, preserving text clarity, and improving OCR accuracy.

4. Proposed Method

Distortions in document images create irregular structures, altering the natural planar arrangement of text. These deformed patterns serve as cues for rectification. Our proposed framework, illustrated in Figure 2, effectively corrects various geometric distortions through an automated learning process. The proposed method utilizes a Stable Diffusion Transformer [19] framework to rectify distorted document images. The approach integrates a Vision Transformer (ViT) for feature extraction, a lightweight segmentation module for foreground extraction, a Thin Plate Spline (TPS) transformation for spatial alignment, and a Conditional Stable Diffusion Transformer Module (CSDTM) for high-quality rectification. Below, we describe our proposed method in detail.

The input distorted document image $I_d \in \mathbb{R}^{H \times W \times 3}$ is first divided into non-overlapping patches of size $P \times P$, forming a sequence of patch embeddings.

$$X_p = \{x_1, x_2, \dots, x_N\}, \quad x_i \in \mathbb{R}^{P^2 \cdot C}$$

Where, $N = \frac{H \times W}{P^2}$ is the number of patches, and C is the number of channels. Each patch embedding is linearly projected into a latent space and positional encodings E_p are added to retain spatial information.

$$Z_0 = X_p + E_p$$

This encoded representation is passed through multiple layers of Multi-Head Self-Attention (MHSA) and Feed-Forward Networks (FFN).

$$Z^{l+1} = \text{MHSA}(\text{LN}(Z^l)) + Z^l$$

$$Z^{l+2} = \text{FFN}(\text{LN}(Z^{l+1})) + Z^{l+1}$$

Where, LN represents Layer Normalization, and l indexes the ViT layers. The extracted features from the last ViT block encode rich global and local document features. These features are then passed through the convolutional layers. Simultaneously, the input image I_d is

passed through a lightweight segmentation module [20] that removes background clutter while preserving textual and structural content. This process outputs the foreground document mask M_f .

$$M_f = \mathcal{S}(I_d)$$

Where, $\mathcal{S}$ represents the segmentation function. The segmented document image is obtained as:

$$I_f = I_d \odot M_f$$

Where, $\odot$ denotes element-wise multiplication. To align the document, a control point detection module identifies key points $P_c = \{p_1, p_2, \dots, p_K\}$ on the document boundary. A Thin Plate Spline (TPS) transformation $\mathcal{T}$ [1] is then applied to warp the document towards a canonical shape.

$$I_w = \mathcal{T}(I_f, P_c)$$

Where, I_w is the warped document. The warped document I_w is encoded using a CLIP-based encoder-decoder framework:

$$F_c = \mathcal{E}(I_w)$$

$$I_u = \mathcal{D}(F_c)$$

Where, $\mathcal{E}$ is the CLIP image encoder, $\mathcal{D}$ is the upsampling decoder. This intermediate representation captures rectified document structures at multiple scales. This is then passed through the convolutional layers. Finally, a cross-attention mechanism is used to merge feature maps from previous convolutional layers with this refined representation.

$$I_c = \text{CrossAttention}(I_u, Z^{l+2})$$

Where, I_c serves as the condition passed to CSDTM for the rectification of the distorted image. For the final rectification, we pass the background-subtracted image I_f and condition I_c into the Conditional Stable Diffusion Transformer Module (CSDTM). At each denoising step t , the reverse diffusion process updates the latent representation using a noise prediction network. The diffusion process progressively removes noise, reconstructing a high-quality rectified image I_r . By leveraging cross-attention, the conditioned information is seamlessly integrated into the transformer's architecture within the diffusion model. This mechanism enables the model to concentrate on specific regions or features in the image that align with the provided conditioning information. With these enriched embeddings, the diffusion model can effectively refine and rectify distorted document images while preserving crucial structural details. This approach significantly enhances the model's ability to address the intricate challenges associated with document image distortion correction.

4.1. Conditional Stable Diffusion Transformer Module (CSDTM)

The Conditional Stable Diffusion Transformer Module (CSDTM), as shown in Figure 3, is designed to perform document image rectification by leveraging a transformer-based architecture instead of the conventional UNet used in standard Stable Diffusion models. Unlike the typical UNet-based approach, which relies heavily on convolutional operations for feature extraction and denoising, our transformer-based framework offers a global receptive field, making it more effective for handling complex geometric distortions in document images. The self-attention mechanism in transformers allows the model to capture long-range dependencies, preserving both local and global document structures, which is critical for accurate rectification. The diffusion process in our model follows a standard forward and reverse mechanism, where noise is gradually added to the distorted document image I_d and later removed in a controlled manner to generate the rectified output I_r . The forward process introduces Gaussian noise at each step t .

$$q(I_t | I_{t-1}) = \mathcal{N}(I_t; \sqrt{\alpha_t} I_{t-1}, (1 - \alpha_t) \mathbf{I})$$

Where, α_t is a noise scheduling parameter. In contrast to the conventional UNet-based denoising network, which processes images in a hierarchical manner using convolutional downsampling and upsampling, our model utilizes a vision transformer (ViT) backbone to extract multi-scale features from the input image. The reverse denoising process is then parameterized as,

$$p_\theta(I_{t-1} | I_t, I_c) = \mathcal{N}(I_{t-1}; \mu_\theta(I_t, I_c, t), \sigma_\theta^2 \mathbf{I})$$

Where, I_c represents the conditional information derived from auxiliary rectification modules such as background segmentation, feature encoding, and control point estimation. A core advantage of our transformer-based architecture is its ability to retain structural consistency through multi-head self-attention. Instead of using convolutional filters that focus on local features, transformers encode global dependencies across the entire document. Given an input

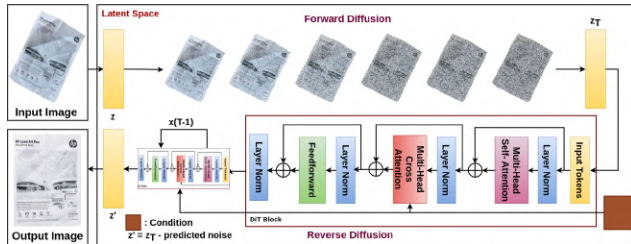


Figure 3. Proposed architecture of Conditional Stable Diffusion Transformer Module (CSDTM).

document image divided into patches, self-attention is computed as,

$$Q = W_q X, \quad K = W_k X, \quad V = W_v X$$

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V$$

Where, X represents the extracted feature embeddings, and d is the dimensionality of each attention head. This formulation allows the transformer model to correlate distant regions of the image, ensuring that the rectified output maintains document integrity across the entire layout.

To effectively guide the rectification process, cross-attention is introduced to inject conditional information I_c directly into the transformer-based denoising network. Unlike UNet, where conditions are typically concatenated with feature maps at different resolutions, cross-attention in our model explicitly aligns the distorted input with structured document priors. Given query features from the distorted image and key-value pairs from the conditioning information, cross-attention is formulated as:

$$Q = W_q I_t, \quad K = W_k I_c, \quad V = W_v I_c$$

$$\text{Cross-Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V$$

This ensures that the rectification process remains aware of background-subtracted document structures, improving text alignment and geometric consistency. After denoising, the output of the transformer is progressively upsampled using convolutional layers. Unlike UNet, where deconvolution layers often introduce artifacts, transformers preserve fine-grained details more effectively. The final rectified image I_r is generated through:

$$I_r = p_\theta(I_0 | I_d, I_c)$$

Where, I_0 represents the estimated clean document obtained through the reverse diffusion process. The shift from UNet to a transformer-based diffusion model brings several benefits, particularly for document image rectification. Self-attention allows the model to understand global distortions rather than focusing only on local pixel relationships. Cross-attention ensures that documents are rectified with respect to their original layout, preserving text alignment. Unlike convolutional architectures that struggle with severe non-uniform distortions, transformers adaptively learn transformations required for rectification.

The Conditional Stable Diffusion Transformer Module (CSDTM) presents a novel approach to document image

rectification by combining transformer-based feature extraction, self-attention mechanisms, and diffusion-based generative modeling. By replacing UNet with a transformer, we achieve a more flexible, context-aware rectification framework that generalizes well across diverse document distortions. This formulation ensures that our model can accurately restore document structure, outperforming traditional rectification approaches.

5. Dataset Details

To train and evaluate our document image rectification model, we utilized several widely recognized datasets that encompass diverse document distortions, real-world challenges, and ground truth annotations. The selected datasets include Doc3D, DocUNet Benchmark, DIR300, and UVDoc, each contributing unique characteristics essential for assessing rectification performance.

The Doc3D dataset is one of the most comprehensive datasets for document unwarping, specifically designed for three-dimensional document rectification. It consists of 100,000 synthetically warped document images, created using real-world document data. Each distorted image is accompanied by ground truth annotations, including 3D coordinate maps, depth maps, and warping flow information, enabling precise learning of distortion correction patterns. This dataset plays a crucial role in model training by offering a large-scale, diverse set of simulated deformations, ensuring robust learning across various document structures. The DocUNet Benchmark dataset is primarily used for evaluation rather than training. It comprises 130 real-world document images, captured from 65 distinct documents, providing a realistic benchmark for assessing model performance. Unlike Doc3D, which contains synthetically generated distortions, DocUNet Benchmark images are captured in real-world settings, making it an important resource for validating rectification models under practical conditions. The DIR300 dataset introduces additional real-world complexity by incorporating 300 document images captured using a mobile phone. Unlike the synthetic datasets, DIR300 reflects diverse background conditions, including varying illumination, occlusions, and perspective distortions. These variations make it a valuable dataset for testing the generalization capability of rectification models in unconstrained environments. The UVDoc dataset offers pseudo-photorealistic document images with accurate 3D shape annotations and unwarping function labels. This dataset includes a broad spectrum of document types, such as printed and handwritten text, structured forms, and diagrams, ensuring versatility in rectification tasks. The images are captured under real-world conditions, incorporating different lighting scenarios, backgrounds, and environmental distortions, further strengthening the model’s ability to handle diverse document layouts and distortions.

By leveraging these datasets, our approach ensures a comprehensive evaluation of our proposed document rectification models across synthetic, real-world, and challenging scenarios. The combination of Doc3D for large-scale training, DocUNet for evaluation, DIR300 for real-world variability, and UVDoc for diverse document types provides a well-rounded benchmark for assessing the effectiveness of document rectification techniques.

6. Experiments

6.1. Metrics

To comprehensively evaluate the effectiveness of our proposed approach, we employed multiple quantitative metrics. These include Edit Distance (ED) to measure text-based discrepancies, Local Distortion (LD) to assess geometric inaccuracies, Multi-Scale Structural Similarity (MS-SSIM) to capture perceptual similarities across different resolutions, Aligned Distortion (AD) to evaluate spatial misalignment, and Character Error Rate (CER) to quantify OCR performance on rectified document images.

6.2. Implementation Details

Our proposed method was implemented using the PyTorch framework. The diffusion transformer uses the ground truth image during training and the input image while inference. The training process follows the typical denoising diffusion probabilistic model (DDPM) framework [21], where a forward process gradually adds noise to the input image, and a reverse process learns to remove the noise to reconstruct the original image. During training, we utilized a batch size of 40 and optimized the model using the Adam optimizer with an initial learning rate of 1×10^{-3} . The learning rate was gradually reduced by a factor of 0.1 after 20 epochs to facilitate stable convergence. The training process spanned 60 epochs and was conducted on two NVIDIA GTX 3090 Ti GPUs. For text recognition tasks, we employed Tesseract OCR (version v5.0.1), as its effectiveness has been vali-



Figure 4. Qualitative comparison of our proposed work with existing methods, including DocProj [15], DewarpNet [5], DocGeoNet [8], DocTr [7], PaperEdge [18], and DocTr++ [10].

dated in previous studies [7, 9].

6.3. Qualitative Results

Figure 4 provides a qualitative comparison of our proposed model against existing state-of-the-art methods. This evaluation was carried out using the DocUNet benchmark dataset, ensuring that all methods were trained on identical samples for a fair comparison. The results demonstrate that our approach consistently achieves superior rectification performance compared to other techniques. One notable limitation of the DocUNet Benchmark dataset [17] is that many of its training samples contain minimal background areas. As a result, models trained on this dataset often struggle when applied to test images with significantly larger background regions, leading to suboptimal rectification. To further examine this issue, we captured a diverse set of document images featuring extensive background regions. The rectification results, illustrated in Figure 5, highlight that our proposed method effectively preserves document integrity while accurately handling broader background areas, thereby outperforming existing approaches in such challenging scenarios. Additionally, our method demonstrates robust performance in rectifying handwritten documents, a task that is inherently more complex due to variations in writing styles, ink intensity, and spatial distortions. As depicted in Figure 5, our approach successfully reconstructs and aligns handwritten text with minimal distortion, further reinforcing its effectiveness across a wide range of document types.

6.4. Quantitative Results

To comprehensively evaluate the effectiveness of our proposed model, we conducted a rigorous quantitative analysis by comparing its performance against existing state-of-the-art. The results of this comparative study on the DocUNet Benchmark dataset [17] are summarized in Table

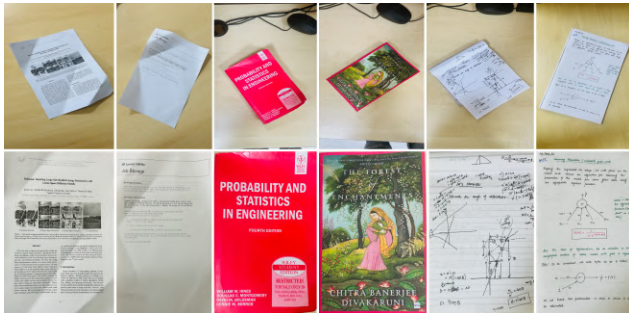


Figure 5. Qualitative results showcasing the effectiveness of our proposed method on handwritten document images and document images with wide backgrounds captured using a mobile phone. The corresponding rectified outputs demonstrate the model’s ability to restore document structure and alignment.

Table 1. Quantitative evaluation of our proposed method in comparison with existing approaches on the DocUNet Benchmark dataset. “↑” indicates higher values are better, while “↓” signifies lower values are preferable.

Methods	MS-SSIM ↑	LD ↓	ED ↓	CER ↓
Distorted	0.25	20.51	1552.22	0.5089
DocUNet [17]	0.41	14.19	1259.83	0.3966
DocProj [15]	0.29	18.01	1165.93	0.3818
FCN-based [25]	0.45	7.84	1031.40	0.3156
DewarpNet [5]	0.47	8.39	525.45	0.2102
DocTr [7]	0.51	7.76	464.83	0.1746
PWUNet [6]	0.49	8.64	743.32	0.2623
DDCP [26]	0.47	8.99	745.35	0.2626
Marior [28]	0.48	7.44	593.80	0.2136
RDGR [11]	0.50	8.51	420.25	0.1559
DocGeoNet [8]	0.50	7.71	379.00	0.1509
DocTr++ [10]	0.51	7.52	447.47	0.1695
CGU-Net [23]	0.51	7.18	689.54	0.1840
MataDoc [4]	0.50	7.42	392.48	0.1505
Foreground and Text [13]	0.49	8.43	697.52	0.1705
LA-DocFlatten [14]	0.52	6.72	391.90	0.1530
DRNet [16]	0.51	7.42	644.48	0.1649
UVDoc [24]	0.54	6.83	707.00	0.1720
PolarDoc [30]	0.49	7.24	868.88	0.2460
DocTLNet [12]	0.51	6.70	377.12	0.1504
DocReal [27]	0.50	7.03	359.90	0.1441
Ours	0.52	6.68	359.00	0.1439

1. Our findings reveal that our approach consistently surpasses prior methods, achieving superior rectification accuracy across multiple evaluation metrics.

In addition to assessing performance on the DocUNet dataset, we extended our evaluation to the DIR300 dataset [9], which contains more diverse and challenging real-world document images. As presented in Table 2, our method outperforms recent state-of-the-art techniques across all quantitative metrics, demonstrating its robustness and adaptability to varying document distortions. Notably, all methods in this comparison were trained on the same dataset, ensuring a fair and unbiased evaluation. Furthermore, we evaluated our model’s effectiveness using the UVDoc dataset [24], a collection of pseudo-photorealistic document images with detailed 3D annotations. The comparative results, detailed in Table 3, further reinforce the superiority of our approach. Our method not only achieves the best performance on the UVDoc dataset but also consistently outperforms existing techniques across all datasets examined. This extensive quantitative analysis underscores the generalizability and efficacy of our model in handling a wide range of document distortions, validating its position as a state-of-the-art solution for document image rectification.

7. Ablation Study

To thoroughly evaluate the significance of each module in our proposed architecture, we conducted an ablation study using the DocUNet dataset [17]. This study systematically examines the contribution of different architectural compo-

Table 2. Quantitative evaluation of our proposed method against existing approaches on the DIR300 test set. For different metrics “↑” shows that the higher the value better the result, and “↓” represents the lower the better.

Methods	MS-SSIM ↑	LD ↓	ED ↓	CER ↓
Distorted	0.3169	39.58	1500.56	0.5234
DocProj [15]	0.3246	30.63	958.89	0.3540
DewarpNet [5]	0.4921	13.94	1059.57	0.3557
DocTr [7]	0.6160	7.21	699.63	0.2237
DDCP [26] 1	0.5524	10.95	2084.97	0.5410
DocGeoNet [8]	0.6380	6.40	664.96	0.2189
Foreground ... [13]	0.6070	7.68	652.80	0.2115
MataDoc [4]	0.6389	5.75	482.90	0.1785
PolarDoc [30]	0.6050	7.17	549.64	0.1680
Ours	0.6458	5.63	482.24	0.1692

Table 3. Quantitative evaluation of our proposed method against existing approaches on the UVDoc dataset. “↑” signifies that a higher value corresponds to better results, while “↓” denotes that a lower value is preferable.

Methods	MS-SSIM ↑	AD ↓	ED ↓	CER ↓
DewarpNet [5]	0.600	0.189	338	0.115
DocTr [7]	0.684	0.176	192	0.065
DDCP [26]	0.591	0.334	362	0.117
RDGR [11]	0.603	0.314	190	0.065
DocGeoNet [8]	0.714	0.167	196	0.066
UVDoc [24]	0.784	0.122	202	0.072
Ours	0.800	0.120	186	0.063

Table 4. Ablation study of the proposed architecture on the DocUNet Benchmark dataset. TPS refers to Thin Plate Spline, CLIP E/D represents the CLIP image encoder-decoder, and CSDTM denotes the Conditional Stable Diffusion Transformer Module.

	MS-SSIM ↑	LD ↓	ED ↓	CER ↓
TPS warping	0.32	17.21	1123.29	0.3138
CLIP E/D	0.40	14.28	1015.01	0.2916
CSDTM	0.29	18.38	1182.93	0.3871
TPS warping + CLIP E/D	0.42	9.82	632.12	0.2561
TPS warping + CSDTM	0.48	7.46	417.32	0.1714
CLIP E/D + CSDTM	0.50	7.32	408.21	0.1627
TPS warping + CLIP E/D + CSDTM	0.52	6.68	359.00	0.1439

nents by analyzing their impact on overall performance. We experimented with multiple model variants, each incorporating different combinations of our network’s core modules, as detailed in Table 4. The results of this study clearly demonstrate the indispensability of each module, as removing any component led to a noticeable decline in performance. These findings highlight that the integration of all modules yields the highest accuracy, reinforcing the effectiveness of our complete model in achieving state-of-the-art (SOTA) performance.

Beyond accuracy, we also examined the computational efficiency of our framework by comparing it with other leading SOTA techniques under a consistent experimental setup. The inference time was measured under identical

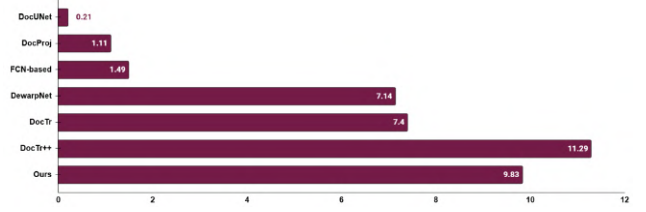


Figure 6. Comparison of the runtime efficiency of our proposed method with state-of-the-art approaches, measured in frames per second (FPS).

conditions to ensure a fair comparison. The results, illustrated in Figure 6, reveal that our approach not only outperforms existing methods in terms of accuracy but also maintains a competitive inference speed. Although our method does not achieve the fastest inference time among all approaches, it maintains a competitive processing speed that is well-suited for practical applications. The balance between efficiency and accuracy ensures that our model delivers high-quality rectification results without significantly compromising runtime performance. This makes our approach feasible for real-world scenarios where both precision and reasonable inference time are essential.

8. Conclusion

In this work, we have introduced a novel Conditional Stable Diffusion Transformer based framework for document image rectification. Our approach effectively leverages self-attention mechanisms to capture both local and global dependencies, making it highly suitable for handling complex geometric distortions in document images. Unlike traditional convolution-based rectification methods, which struggle with non-uniform deformations, our transformer-based diffusion model provides a context-aware and structure preserving rectification process. Through extensive experiments on multiple benchmark datasets, including DocUNet, DIR300, and UVDoc, we demonstrated that our proposed method achieves state-of-the-art performance across various evaluation metrics. Furthermore, our ablation study confirmed the critical role of each module in enhancing the network’s overall effectiveness. By integrating transformer-based feature extraction, cross-attention conditioning, and diffusion-based generative modeling, our framework represents a significant advancement in document image rectification. Future work could explore further optimizations, such as lightweight transformer variants for improved efficiency and multi-modal conditioning to enhance generalizability across diverse document types. This research paves the way for more robust and accurate document restoration techniques, benefiting real-world applications such as OCR systems, archival digitization, and automated document processing.

References

- [1] Adrien Bartoli, Mathieu Perriollat, and Sylvie Chambon. Generalized thin-plate spline warps. *International Journal of Computer Vision*, 88:85–110, 2010. [4](#)
- [2] Michael S Brown and W Brent Seales. Document restoration using 3d shape: a general deskewing algorithm for arbitrarily warped documents. In *Proceedings Eighth IEEE International Conference on Computer Vision. ICCV 2001*, pages 367–374. IEEE, 2001. [2](#)
- [3] Frederic Courteille, Alain Crouzil, J-D Durou, and Pierre Gurdjos. Towards shape from shading under realistic photographic conditions. In *Proceedings of the 17th International Conference on Pattern Recognition, 2004. ICPR 2004.*, pages 277–280. IEEE, 2004. [2](#)
- [4] Beiya Dai, Qunyi Xie, Yulin Li, Xiameng Qin, Chengquan Zhang, Kun Yao, Junyu Han, et al. Matedoc: Margin and text aware document dewarping for arbitrary boundary. *arXiv preprint arXiv:2307.12571*, 2023. [7](#), [8](#)
- [5] Sagnik Das, Ke Ma, Zhixin Shu, Dimitris Samaras, and Roy Shilkrot. Dewarpnet: Single-image document unwarping with stacked 3d and 2d regression networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 131–140, 2019. [2](#), [6](#), [7](#), [8](#)
- [6] Sagnik Das, Kunwar Yashraj Singh, Jon Wu, Erhan Bas, Vijay Mahadevan, Rahul Bhotika, and Dimitris Samaras. End-to-end piece-wise unwarping of document images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4268–4277, 2021. [7](#)
- [7] Hao Feng, Yuechen Wang, Wengang Zhou, Jiajun Deng, and Houqiang Li. Doctr: Document image transformer for geometric unwarping and illumination correction. *arXiv preprint arXiv:2110.12942*, 2021. [2](#), [6](#), [7](#), [8](#)
- [8] Hao Feng, Wengang Zhou, Jiajun Deng, Yuechen Wang, and Houqiang Li. Geometric representation learning for document image rectification. In *Computer Vision – ECCV 2022*, pages 475–492, Cham, 2022. Springer Nature Switzerland. [6](#), [7](#), [8](#)
- [9] Hao Feng, Wengang Zhou, Jiajun Deng, Yuechen Wang, and Houqiang Li. Geometric representation learning for document image rectification. In *Computer Vision–ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23–27, 2022, Proceedings, Part XXXVII*, pages 475–492. Springer, 2022. [7](#)
- [10] Hao Feng, Shaokai Liu, Jiajun Deng, Wengang Zhou, and Houqiang Li. Deep unrestricted document image rectification. *arXiv preprint arXiv:2304.08796*, 2023. [2](#), [6](#), [7](#)
- [11] Xiangwei Jiang, Rujiao Long, Nan Xue, Zhibo Yang, Cong Yao, and Gui-Song Xia. Revisiting document image dewarping by grid regularization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4543–4552, 2022. [7](#), [8](#)
- [12] Pooja Kumari and Sukhendu Das. Am i readable? transfer learning based document image rectification. *International Journal on Document Analysis and Recognition (IJDAR)*, pages 1–14, 2024. [7](#)
- [13] Heng Li, Xiangping Wu, Qingcai Chen, and Qianjin Xiang. Foreground and text-lines aware document image rectification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 19574–19583, 2023. [7](#), [8](#)
- [14] Pu Li, Weize Quan, Jianwei Guo, and Dong-Ming Yan. Layout-aware single-image document flattening. *ACM Transactions on Graphics*, 2023. [7](#)
- [15] Xiaoyu Li, Bo Zhang, Jing Liao, and Pedro V Sander. Document rectification and illumination correction using a patch-based cnn. *ACM Transactions on Graphics (TOG)*, 38(6):1–11, 2019. [2](#), [6](#), [7](#), [8](#)
- [16] Shaokai Liu, Hao Feng, and Wengang Zhou. Re-thinking supervision in document unwarping: A self-consistent flow-free approach. *IEEE Transactions on Circuits and Systems for Video Technology*, 2023. [7](#)
- [17] Ke Ma, Zhixin Shu, Xue Bai, Jue Wang, and Dimitris Samaras. Docunet: Document image unwarping via a stacked u-net. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 4700–4709, 2018. [2](#), [7](#)
- [18] Ke Ma, Sagnik Das, Zhixin Shu, and Dimitris Samaras. Learning from documents in the wild to improve document unwarping. In *ACM SIGGRAPH 2022 Conference Proceedings*, pages 1–9, 2022. [6](#)
- [19] William S Peebles and Saining Xie. Scalable diffusion models with transformers. 2023 ieee. In *CVF International Conference on Computer Vision (ICCV)*, 2022. [4](#)
- [20] Xuebin Qin, Zichen Zhang, Chenyang Huang, Masood Dehghan, Osmar R Zaiane, and Martin Jagersand. U2-net: Going deeper with nested u-structure for salient object detection. *Pattern Recognition*, 106: 107404, 2020. [4](#)
- [21] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020. [6](#)
- [22] Yau-Chat Tsoi and Michael S Brown. Multi-view document rectification using boundary. In *2007 IEEE Conference on Computer Vision and Pattern Recognition*, pages 1–8. IEEE, 2007. [2](#)

- [23] Floor Verhoeven, Tanguy Magne, and Olga Sorkine-Hornung. Neural document unwarping using coupled grids. *arXiv preprint arXiv:2302.02887*, 2023. [7](#)
- [24] Floor Verhoeven, Tanguy Magne, and Olga Sorkine-Hornung. Uvdoc: Neural grid-based document unwarping. In *SIGGRAPH Asia 2023 Conference Papers*, pages 1–11, 2023. [7](#), [8](#)
- [25] Guo-Wang Xie, Fei Yin, Xu-Yao Zhang, and Cheng-Lin Liu. Dewarping document image by displacement flow estimation with fully convolutional network. In *Document Analysis Systems: 14th IAPR International Workshop, DAS 2020, Wuhan, China, July 26–29, 2020, Proceedings 14*, pages 131–144. Springer, 2020. [7](#)
- [26] Guo-Wang Xie, Fei Yin, Xu-Yao Zhang, and Cheng-Lin Liu. Document dewarping with control points. In *Document Analysis and Recognition–ICDAR 2021: 16th International Conference, Lausanne, Switzerland, September 5–10, 2021, Proceedings, Part I 16*, pages 466–480. Springer, 2021. [7](#), [8](#)
- [27] Fangchen Yu, Yina Xie, Lei Wu, Yafei Wen, Guozhi Wang, Shuai Ren, Xiaoxin Chen, Jianfeng Mao, and Wenye Li. Docreal: Robust document dewarping of real-life images via attention-enhanced control point prediction. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 665–674, 2024. [2](#), [7](#)
- [28] Jiaxin Zhang, Canjie Luo, Lianwen Jin, Fengjun Guo, and Kai Ding. Marior: Margin removal and iterative content rectification for document dewarping in the wild. *arXiv preprint arXiv:2207.11515*, 2022. [7](#)
- [29] Li Zhang, Yu Zhang, and Chew Tan. An improved physically-based method for geometric restoration of distorted document images. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 30(4): 728–734, 2008. [2](#)
- [30] Weiguang Zhang, Qiufeng Wang, and Kaizhu Huang. Polar-doc: One-stage document dewarping with multi-scope constraints under polar representation. *arXiv preprint arXiv:2312.07925*, 2023. [7](#), [8](#)